MEDICAL IMAGE PROCESSING

Master's Degree in Biomedical Engineering

Liver Steatosis Challenge

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1. Introduction

Hepatic steatosis is a condition characterized by the intracellular accumulation of triglycerides in the form of lipid vacuoles. In the transplant setting, the accurate assessment of graft quality is increasingly critical due to the evolving definition of Metabolic Dysfunction–Associated Steatotic Liver Disease (MASLD) [1] and its rising global burden [2]. Indeed, a high degree of steatosis is associated with an increased risk of rejection and primary organ dysfunction; therefore, organ screening and selection require an accurate and reproducible assessment.

The current gold standard is the histopathological analysis of biopsies stained with hematoxylin and eosin (H&E). However, this evaluation is affected by intra- and inter-observer variability and is influenced by acquisition artefacts and the histological heterogeneity of steatosis [3]. Moreover, the increasing prevalence of steatotic livers has led to a growing workload for pre-transplant evaluation centers.

To address these limitations, this work proposes an automated pipeline based on convolutional neural networks (CNNs) for the segmentation of lipid vacuoles [3,5], producing a binary output (background vs steatosis) and evaluated according to the metrics defined by the challenge.

2. Materials and Methods

2.1 Dataset Composition and Partitioning

The initial dataset consists of 510 histological images of hepatic steatosis, each accompanied by a manual segmentation performed by an expert operator (Fig. 1).

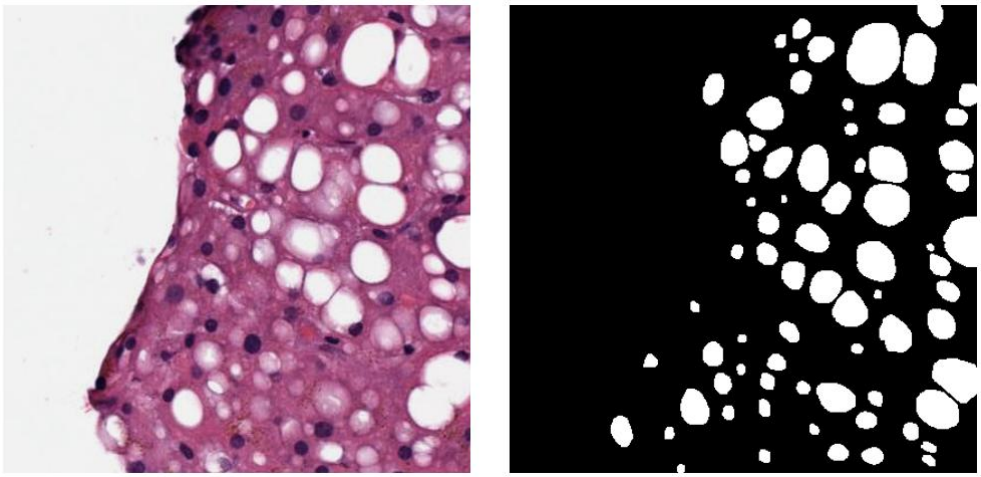
(a)  (b)

Fig. 1 - Example of histological image of hepatic steatosis (a) with the corresponding manually segmented mask

The original data split provided includes 454 images for the Training Set (TRS) and 56 images for the Validation Set (VS). To obtain an independent test set of comparable size to the validation set, a Test Set (TS) consisting of 56 images was randomly extracted from the original TRS. This results in a practical data partition of approximately 80% TRS - 10% VS - 10% TS, with VS and TS having equal cardinality, which is consistent with the objective of enabling a stable and balanced comparison between validation and test phases.

2.2 Metrics for Evaluating Results

To evaluate the performance of the automatic steatosis segmentation algorithm, the predicted binary masks were compared with the corresponding reference manual annotations provided in the dataset (i.e., Ground Truth, GT). All evaluation metrics were computed on a per-image basis, and results are reported as mean ± standard deviation (SD) over the entire test set.

2. MATERIALS AND METHODS

In accordance with the challenge guidelines, three main evaluation metrics were considered. The first metric is the Dice Similarity Coefficient (DSC), which quantifies the spatial overlap between automatic and manual segmentation masks and is defined as (Eq. 1):

$$DSC(X,Y) = 2 \cdot |X \cap Y| / (|X| + |Y|) \tag{1}$$

where X represents the manual mask and Y the automatic mask.

Since the challenge focuses on the segmentation of individual steatosis vacuoles, the second metric adopted was Count Error. The number of vacuoles was estimated as the number of connected components in each binary mask, and the counting error was defined as (Eq. 2):

$$Count_{error} = |N_{manual} - N_{automatic}| \tag{2}$$

where N_{manual} and $N_{automatic}$ denote the number of steatosis vacuoles identified in the manual and automatic segmentations, respectively.

The third and final primary metric considered was the steatosis percentage error, aimed at evaluating the accuracy of global steatosis quantification. For each image, the steatosis percentage was defined as (Eq. 3):

$$SP = (Area_{steatosis} \,/\, Area_{image}) \cdot 100 \tag{3}$$

The absolute steatosis percentage error was then defined as (Eq. 4):

$$Steatosis\ Error = |SP_{manual} - SP_{automatic}| \tag{4}$$

To further characterize segmentation performance, additional complementary metrics were computed. In particular, the Intersection over Union (IoU) was evaluated to measure the geometric overlap between the predicted mask and the GT, penalizing both over-segmentation and under-segmentation. This metric is particularly informative for assessing spatial segmentation quality and is defined as (Eq. 5):

$$IoU = |X \cap Y| \,/\, (X \cup Y) = TP/(TP + FP + FN) \tag{5}$$

Precision was also evaluated to measure how often positive predictions are correct, making it useful for quantifying the tendency of the model to generate false positives together with pixel-wise recall, which measures how many true vacuoles are detected. Recall is crucial for assessing false negatives, which lead to an underestimation of the steatotic burden and to incomplete or perforated segmentation masks. These two metrics are defined as (Eq. 6, 7):

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

where TP, TN, FP, and FN represent True Positive, True Negative, False Positive, and False Negative, respectively.

For completeness, pixel-wise accuracy was also reported, indicating the overall proportion of correctly classified pixels and defined as (Eq. 8):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

although its interpretability is limited by the strong class imbalance between the background and steatotic regions.

Finally, Hausdorff Distance (HD) was computed to assess boundary discrepancies between manual and automatic segmentations. Using the symmetric definition, it is given by (Eq. 9):

$$d_H(X,Y) = \max\left\{\sup_{x\in X}\inf_{y\in Y} d\,(x,y),\ \sup_{y\in Y}\inf_{x\in X} d\,(x,y)\right\} \tag{9}$$

Metrics were computed on a per-image basis; in empty–empty cases (both GT and prediction empty), Dice and IoU are undefined (0/0) and were therefore recorded as NaN (Not a Number). Final statistics (mean ± SD) were computed using *nanmean/nanstd*, also reporting the number of NaN values. In addition, micro-averaged metrics were provided by aggregating TP/FP/FN/TN to obtain global performance indicators over the entire dataset.

3. Pre-Processing

Histological liver images exhibit chromatic and structural variability related to staining procedures, scanners, and acquisition conditions (Fig. 2).

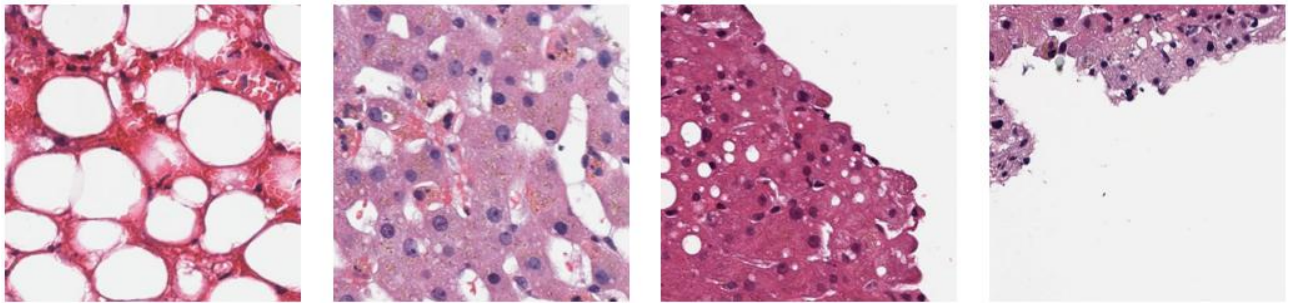


Fig. 2 - Four different examples of hepatic steatosis histological images highlighting their high variability

To reduce non-biological variability and limit the risk of domain shift, a preprocessing pipeline was initially evaluated, consisting of:

- Color Normalization (Macenko method),
- Median Filtering,
- Sharpening (Unsharp masking),
- Contrast Limited Adaptive Histogram Equalization (CLAHE),
- Geometric data augmentation applied only to the training set.

3.1 Color Normalization (Macenko Method)

Color normalization was performed using the Macenko stain normalization method [6], with the aim of reducing the chromatic variability typically observed in histological images acquired under heterogeneous conditions (e.g., different laboratories, scanners, staining protocols, fixation times, and staining durations). If left uncontrolled, this variability may introduce domain shift and induce the model to exploit color and intensity differences as non-robust shortcuts, ultimately degrading generalization performance on the TS.

From an operational standpoint, the algorithm first converts the image into the optical density (OD) space, where light absorption is more consistent with the physics of histological staining. Pixels carrying low information content (below a threshold β, typically corresponding to background or noise) are then excluded, and the dominant stain basis is estimated through covariance analysis (PCA/eigen decomposition) of the OD

vectors. The two principal eigenvectors are interpreted as directions associated with the main stains (in practice, hematoxylin and eosin or equivalent components), and robust stain directions are obtained using controlled percentiles (α).

Subsequently, per-pixel stain concentrations are computed via a least-squares solution, and a re-scaling is applied with respect to predefined reference parameters (HERef, maxCRef) to map the image to a standardized color space. Finally, the image is reconstructed and converted back to the RGB domain.

This normalization step was adopted because, in a liver steatosis vacuole segmentation challenge, it is crucial that the network learns morphological and structural patterns (e.g., boundaries, parenchymal texture, vacuole–tissue transitions) rather than incidental color variations. By standardizing stain appearance, the overall visual homogeneity of the dataset is increased, thereby enhancing the likelihood that the model learns more robust and transferable features across images acquired from different sources (Fig. 3).

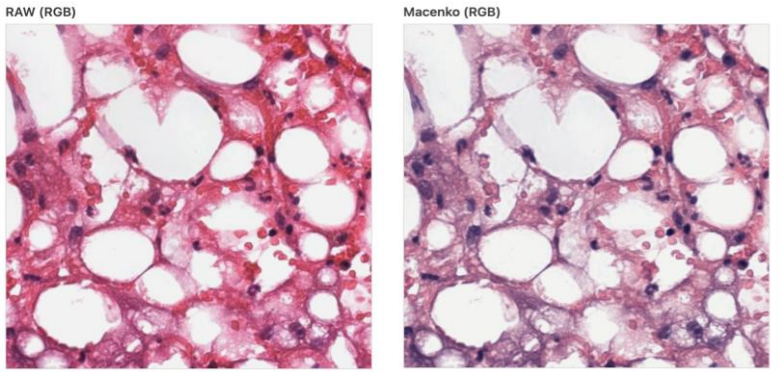


Fig. 3 - Macenko stain normalization effect

3.2 Median Filtering

Median filtering was applied as a local, edge-preserving denoising step. Each pixel value is replaced by the median within a small neighborhood (e.g., 3 × 3), a robust statistic that suppresses impulsive noise and small artefacts with limited edge blurring compared to linear filters. In histological images, this reduces acquisition/preparation micro-artefacts and limits the propagation of non-informative high-frequency variations to downstream steps, improving input stability and segmentation robustness while preserving vacuole–tissue boundaries (Fig. 4).

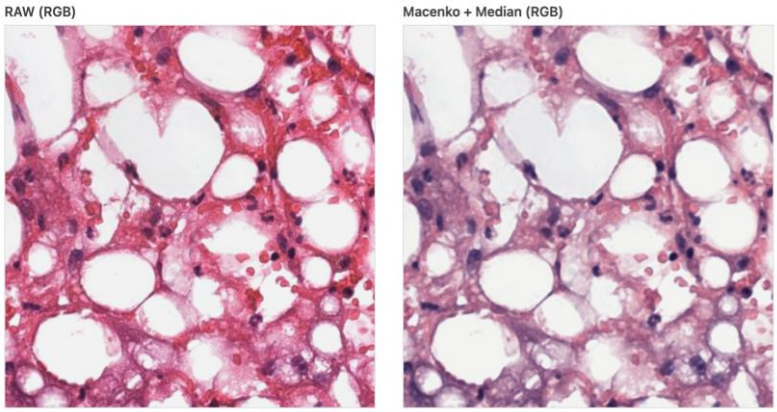


Fig. 4 - Macenko stain normalization + Median filter effect

3.3 Sharpening

Sharpening was implemented via Gaussian-based unsharp masking. A blurred (low-frequency) version of the image is first computed, then the amplified residual between the original and blurred images is added back through a weighted combination, enhancing edges and fine details. This step was introduced to improve the

definition of vacuole boundaries, which can be weak due to acquisition, preparation, and staining variability, thereby facilitating boundary learning by the segmentation model. Parameters (blur σ and strength) were tuned to limit over-sharpening artefacts (e.g., halos and noise amplification) that could introduce spurious edges and harm segmentation performance (Fig. 5).

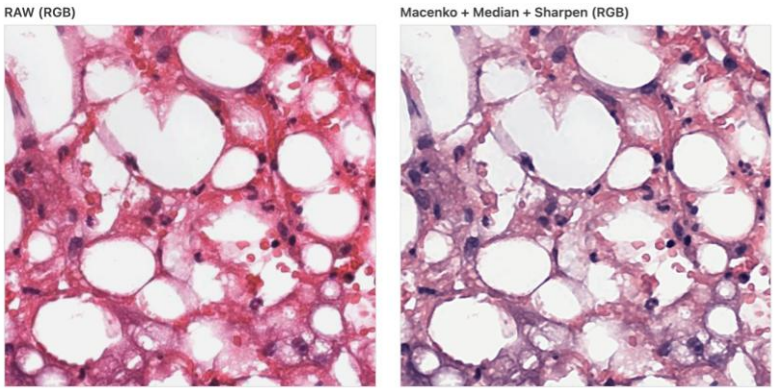


Fig. 5 - Macenko stain normalization + Median filter + Sharpening effect

3.4 CLAHE

CLAHE (Contrast Limited Adaptive Histogram Equalization) was used to enhance local contrast and to make structures and intensity transitions more distinguishable, which in histological images may appear weak or non-uniform. Operationally, the image is converted to grayscale and divided into small regions (tiles, defined by the tile_grid_size). For each tile, histogram equalization is applied with a contrast-limiting constraint (clip_limit): the histogram is clipped to prevent excessive noise amplification, and the final image is obtained by interpolating between neighboring tiles to preserve spatial continuity.

The use of CLAHE is motivated by the fact that, in the segmentation of vacuoles in liver tissue, the discrimination between bright areas (vacuoles/lipid spaces) and the surrounding tissue often relies on local intensity differences rather than on a consistent global contrast. Moreover, non-uniform illumination, residual staining variability, and acquisition differences may lead to regions with reduced contrast. CLAHE helps locally normalize the grayscale dynamic range, making task-relevant details more stable and comparable across different images (Fig. 6).

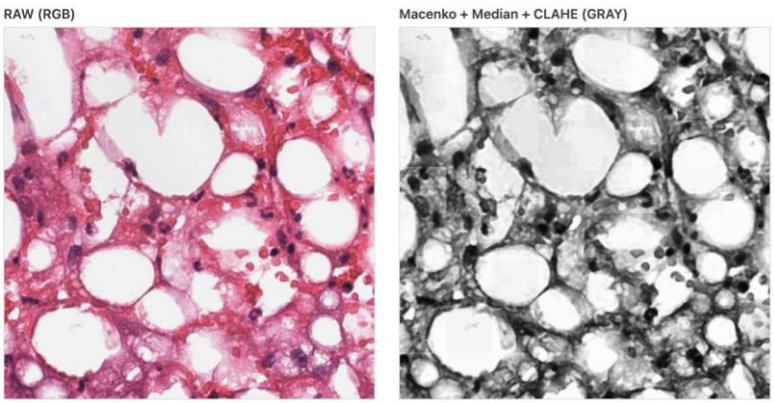


Fig. 6 - Graphical demonstration of the entire pipeline effect

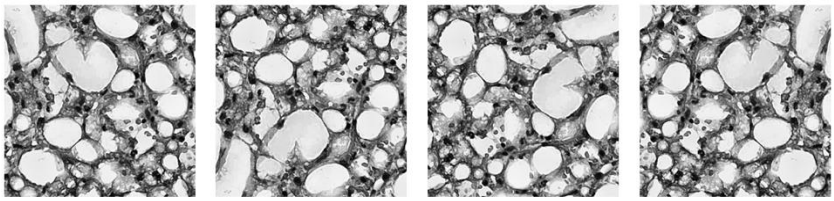
3.4.1 Implementation note

In the initial implementation, CLAHE was applied to the entire image after conversion to grayscale, with the aim of increasing local contrast and making vacuole–parenchyma transitions more evident. In hindsight, this choice proved to be suboptimal, as grayscale conversion compresses the chromatic information of H&E staining into a single intensity map, effectively reducing the benefit of Macenko normalization, which is specifically designed to standardize staining components and mitigate color-related domain shift. Moreover,

emphasizing local contrast in a single-channel representation may amplify non-biological variations (e.g., texture and micro-artefacts), introducing less robust signals for the model.

3.5 Data Augmentation

Geometric data augmentation was applied only to the TRS to improve robustness and reduce overfitting. For each sample, one label-preserving transformation was randomly selected (horizontal/vertical flip, 90° rotation, or identity) and applied identically to the image and its masks to maintain perfect alignment. Three augmented variants per image were generated, expanding TRS and promoting rotation/reflection invariance, which is appropriate for vacuole segmentation where orientation is not informative (Fig. 7).



Pre_processed Aug_vertical Rotate_90 Aug_horizontal

Fig. 7 - Augmented image

3.6 Pre-Processing Ablation Study

The preprocessing strategy was validated through a dedicated ablation study, aimed at quantifying the contribution of each block and verifying that the final configuration consistently improved the evaluation metrics. The study was implemented as a sequential removal of modules following the pipeline order (leave-one-out), rather than an exhaustive grid search, to avoid combinatorial growth and to keep the comparisons interpretable. By keeping dataset splits and training procedure, performance changes could be attributed specifically to preprocessing. Results were benchmarked against a RAW DATA baseline (augmentation only) and then compared to progressively simplified variants obtained by removing one step at a time.

Compared to the RAW DATA baseline, introducing preprocessing yields a clear improvement, confirming that standardization and filtering provide a tangible benefit (Tab. 1). Removing the median filter degrades performance, indicating that this initial denoising step stabilizes the signal and supports the downstream processing. Conversely, removing sharpening does not harm the results and can be slightly beneficial, suggesting that edge enhancement is not reliably informative and may introduce artefacts. Finally, once sharpening is excluded, the role of CLAHE emerges as a trade-off: keeping it tends to favor overlap/spatial accuracy, whereas removing it shifts the model toward higher sensitivity and improved quantification behavior but with increased over-segmentation (Tab. 1).

ABLATION STUDY Pre-Proc.	RAW DATA	COMPLETE PIPELINE	NO MEDIAN	NO SHARPENING	NO CLAHE & NO SHARPENING
IoU	0.3309 ± 0.2739	0.4193 ± 0.2602	0.3968 ± 0.2730	0.4228 ± 0.2673	0.4074 ± 0.2658
Dice	0.4306 ± 0.3272	0.5378 ± 0.2922	0.5068 ± 0.3170	0.5375 ± 0.3055	0.5209 ± 0.3100
Precision	0.5604 ± 0.3724	0.6963 ± 0.2966	0.6731 ± 0.3287	0.6589 ± 0.3180	0.5769 ± 0.3409
Recall	0.5039 ± 0.2793	0.5343 ± 0.2748	0.5317 ± 0.2741	0.5603 ± 0.2774	0.6291 ± 0.2248
Accuracy	0.9899 ± 0.0146	0.9900 ± 0.0169	0.9902 0.0168	0.9903 ± 0.0157	0.9906 ± 0.0160
HD	134.2758 ± 97.6663	101.7348 ± 65.8785	111.0150 ± 87.6155	101.0134 ± 80.5596	112.6546 ± 90.9829
Count Error	6.7500 ± 0.7485	5.2857 ± 8.8675	5.1071 ± 7.8891	5.2321 ± 7.8898	4.6071 ± 6.1461
Steatosis Error	0.4298 ± 0.7485	0.5159 ± 1.1321	0.5255 ± 1.1315	0.4249 ± 0.8766	0.3999 ± 0.8013

Tab. 1 - results of the pre-processing ablation study

In conclusion, in line with the primary objective of the challenge—accurate vacuole delineation—the **No Sharpening variant (Macenko + median filtering + CLAHE)** was adopted as the final preprocessing configuration (Tab. 1), as it provides the most reliable compromise between spatial segmentation accuracy and boundary stability while avoiding artefact introduction.

4. Neural Networks and Parameters

To ensure controlled comparisons during the early phases of the project, experiments—particularly the preprocessing ablation—were conducted using the network provided in **Laboratory 3** as a baseline and a fixed training protocol, so that performance variations could be attributed exclusively to input transformations.

The reference model is a **2D U-Net** for binary segmentation [7], with an encoder–decoder architecture and skip connections, an architecture widely validated for histological assessment.[4] Inputs were mainly single-channel (in_channels = 1) because the preprocessed images are grayscale; only the "no CLAHE" experiment used a 3-channel input. The network outputs a single logit map (out_channels = 1), converted to probabilities via a sigmoid. Each stage uses DoubleConv blocks (two 3×3 convolutions with padding = 1, each followed by BatchNorm and ReLU). Downsampling is performed with 2×2 max-pooling; with init_filters = 32 and depth = 3, features increase from 32 to 64 to 128, with a 256-channel bottleneck. Upsampling is bilinear (with a 1×1 convolution for channel alignment), followed by concatenation with encoder features and additional DoubleConv blocks; a final 1×1 convolution produces the predicted mask. Images are converted to tensors and normalized to [0,1], while masks are binarized (0 background, 1 steatosis). Training was run for 20 epochs (batch size 4) with Adam (lr = 1×10^{-3}) and a combined loss BCEWithLogits + Dice, pairing stable pixel-wise supervision with an overlap-driven term suited to the strong class imbalance (Fig. 8).

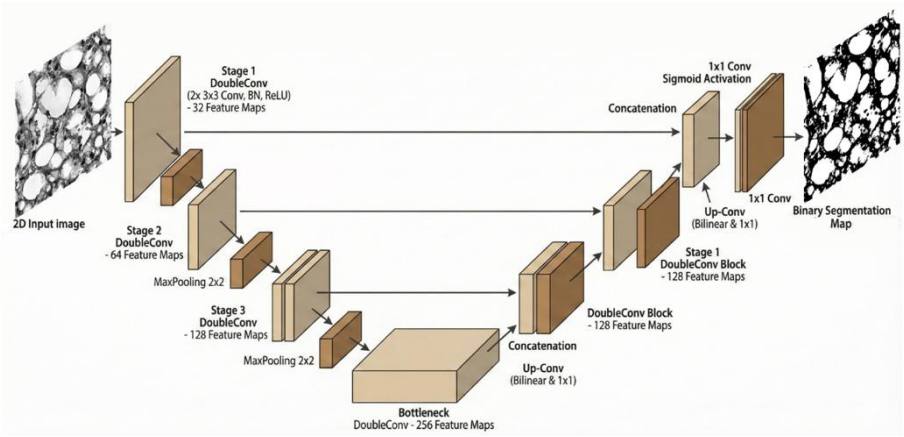


Fig. 8 - graphical representation of the U-Net reference model

4.1 Evaluated architectures (Network Ablation)

A dedicated ablation study on network architectures was subsequently conducted, including eight U-Net–based configurations (Networks 1–7B), varying: bilinear upsampling vs transposed convolutions, Adam vs AdamW optimization, BCE+Dice vs Focal+Dice loss functions, and the inclusion of **Squeeze-and-Excitation (SE)** blocks.[8] (Tab. 2)

Experiment	Modification	Scientific Motivation	Expected Effect
1	init_filters = 64	Increase representational capcity	+ IoU, + Dice
2	depth = 4	Improve multi-scale feature extraction	+ Recall, + precision on details
3	Focal + Dice Loss	Handle class imbalance	+ Recall, improved boundaries
4	AdamW	Greater stability and generalization	- Val Loss, + robustness
5	ConvTranspose2D	More informative decoder	More precise boundaries
6	SE-block into skip	Better attention to tissue	Improved delineation

Tab. 2 - summary table of the six experiments, including their motivation and expected effects

The tested variants were organized as a stepwise ablation to isolate the contribution of architecture and training choices. **Networks 1–2** probe capacity vs depth using bilinear upsampling (64 filters as a high-capacity baseline vs a deeper but narrower model). **Networks 3–4** keep the same structure while changing the training strategy (Focal Loss to handle class imbalance; AdamW with weight decay to improve

regularization). **Networks 5–6** introduce architectural upgrades (learnable upsampling via transposed convolutions; Squeeze-and-Excitation blocks for channel-wise attention). Finally, **Networks 7A–7B** implement the most advanced configuration: **7A (SE-U-Net)** combines higher capacity, transposed convolutions and AdamW with SE modules in skip connections, while **7B** removes SE to quantify the attention benefit. Overall, this systematic design clarifies the impact of each modification and shows that **Network 7A** achieves the best segmentation performance (Tab. 3).

ABLATION STUDY for networks	1	2	3	4	5	6	7A	7B
IoU	0.3992 ± 0.2614	0.3471 ± 0.2640	0.0219 ± 0.0472	0.4162 ± 0.2643	0.4059 ± 0.2591	0.4282 ± 0.2160	0.4602 ± 0.2292	0.3909 ± 0.2643
Dice	0.5142 ± 0.3046	0.4535 ± 0.3181	0.0394 ± 0.0764	0.5308 ± 0.3081	0.5217 ± 0.3048	0.5640 ± 0.2400	0.5902 ± 0.2588	0.5030 ± 0.3138
Precision	0.5799 ± 0.3243	0.4710 ± 0.3286	0.0219 ± 0.0472	0.6082 ± 0.3183	0.5762 ± 0.3278	0.8013 ± 0.1819	0.6716 ± 0.2614	0.5256 ± 0.3357
Recall	0.5880 ± 0.2678	0.6447 ± 0.2614	1.0000 ± 0.0000	0.6072 ± 0.2638	0.6126 ± 0.2493	0.4932 ± 0.2560	0.6135 ± 0.2565	0.6599 ± 0.2425
Accuracy	0.9896 ± 0.0178	0.9894 ± 0.0146	0.0219 ± 0.0472	0.9899 ± 0.0167	0.9898 ± 0.0.179	0.9899 ± 0.0163	0.9909 ± 0.0141	0.9899 ± 0.0167
HD	108.4700 ± 87.9704	115.6655 ± 85.7020	239.9094 ± 129.7771	97.9420 ± 64.7891	100.2702 ± 67.0062	107.4308 ± 76.4492	102.0349 ± 53.5780	109.5423 ± 78.5029
Count Error	5.1250 ± 7.0053	5.9107 ± 8.3351	20.7679 ± 28.6188	5.2321 ± 7.5047	5.0893 ± 6.7540	5.5714 ± 8.6455	4.8929 ± 6.9145	5.1250 ± 6.5522
Steatosis Error	0.4417 ± 1.0442	0.3598 ± 0.6160	97.8061 ± 4.7160	0.3818 ± 0.8319	0.3972 ± 1.1059	0.5936 ± 1.1673	0.3384 ± 0.6173	0.3644 ± 0.9678

Tab. 3 - results of the U-Net ablation study

4.2 Pre-Processing Refinement after Model Selection

Although an initial preprocessing ablation was conducted using a baseline network, those choices are not necessarily optimal once a more expressive architecture is adopted. For this reason, after selecting the final model (SE-U-Net, Network 7A), we re-evaluated the preprocessing blocks while keeping the dataset split, training protocol and thresholding, so that any performance change could be attributed solely to preprocessing.

Removing the median filter improved performance, likely because it denoising effect can suppress fine boundary cues that the SE-U-Net is able to exploit. CLAHE was also excluded: applied after grayscale conversion, it removes H&E chromatic information and may amplify artefacts, without providing stable gains in the main challenge metrics. In conclusion, the **final pipeline** retains ***only Macenko stain normalization as preprocessing***, while mask regularization is delegated to morphological post-processing, yielding a simpler and more robust workflow.

5. Post-Processing

5.1 Threshold Selection and Inference

Model outputs are probability maps that are binarized into vacuole/background masks using a threshold selected on the validation set (VS). We performed a sweep of thresholds from 0.05 to 0.90 (step 0.05), and for each value computed the per-image Dice and then the mean Dice across VS. Images with empty ground truth (no vacuoles) were excluded from this selection to avoid trivial empty predictions. The threshold yielding the highest mean Dice on the non-empty subset was then fixed and applied for inference on the test set (TS). The resulting optimal threshold was ***best_threshold = 0.60***, yielding a mean Dice (VS, non-empty GT) of **0.6909**. The combined effect of the threshold selection and the simplification of pre-processing yielded the following results, showing a clear improvement across all metrics (Tab. 4):

	RAW DATA & 7A	*RAW DATA + 7A + SWEEP THRESHOLD*	*ONLY MACHENKO + 7A + SWEEP THRESHOLD*
IoU	0.4293 ± 0.2567	0.4364 ± 0.2483	0.4878 ± 0.2319
Dice	0.5484 ± 0.2943	0.5597 ± 0.2806	0.6174 ± 0.2489
Precision	0.5358 ± 0.3039	0.6089 ± 0.3143	0.6593 ± 0.2672
Recall	0.7123 ± 0.1983	0.6306 ± 0.2074	0.6652 ± 0.2165
Accuracy	0.9895 ± 0.0151	0.9907 ± 0.0138	0.9917 ± 0.0123
HD	115.9240 ± 88.6812	115.7476 ± 89.0837	101.7944 ± 84.8603
Count Error	7.1964 ± 9.7421	5.4286 ± 6.9869	4.3750 ± 5.5245
Steatosis Error	0.3451 ± 0.6170	0.2662 ± 0.4447	0.2222 ± 0.3260

Tab. 4 - new results with 7A SE-U-Net

5.2 Isolated Evaluation of Morphological Post-Processing

After binarization, a post-processing stage was applied to reduce artefacts and improve mask coherence. The analysis was designed as an isolated ablation study: each operator was applied individually starting from the same initial binary mask, avoiding cumulative effects and enabling a direct comparison. The following operations were evaluated (Fig. 9):

- ***Removal of Small Objects***, discarding tiny, connected components as likely artefacts to reduce isolated false positives and improve counting.
- ***Hole Filling***, closing internal cavities to obtain more compact regions and support area/steatosis quantification.
- ***Morphological Closing (dilation-erosion)***, to smooth boundaries and fill small gaps, with the potential drawback of merging nearby vacuoles
- ***Morphological Opening (erosion–dilation)***, which was introduced as a more conservative operator to suppress small, non-robust detections and thin protrusions while preserving the overall shape of main vacuoles. In practice, the erosion step removes point-like false positives and weak connections produced by texture/contrast fluctuations; the subsequent dilation restores the main structures without reintroducing the spurious components removed during erosion. Importantly, opening can also help **separate adjacent vacuoles** when they are erroneously connected by thin bridges, addressing the common issue of lipid droplet clumping.[5]

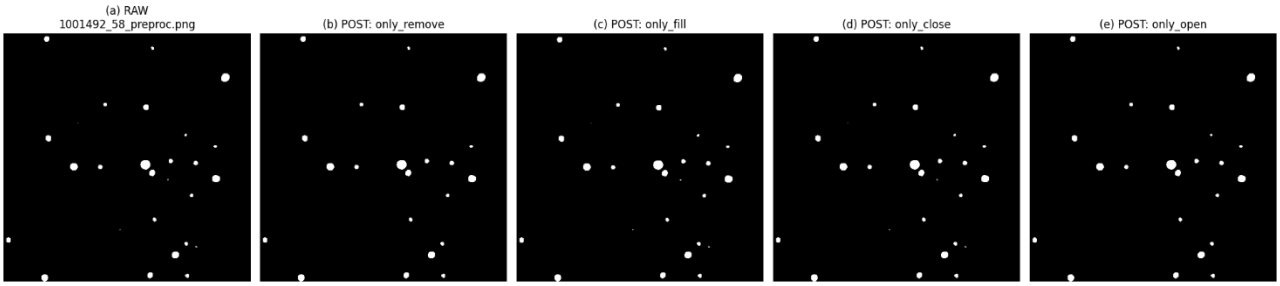


Fig. 9 - graphical examples of the post-processing effect

5.3 Post-Processing Ablation Study

Based on the results (Tab. 5), the selected operator was **morphological opening**, as it can suppress isolated spurious detections while preserving the overall morphology of vacuoles.

	BASELINE no post	*Only REMOVE SMALL OBJECTS*	*Only HOLE FILLING*	*Only CLOSING*	*Only OPENING*
IoU	0.4878 ± 0.2319 | NaN: 19/56	0.4990 ± 0.2223 | NaN: 20/56	0.4878 ± 0.2319 | NaN: 19/56	0.4877 ± 0.2319 | NaN: 19/56	0.5000 ± 0.2203 | NaN: 20/56
Dice	0.6174 ± 0.2489 | NaN: 19/56	0.6316 ± 0.2336 | NaN: 20/56	0.6174 ± 0.2489 | NaN: 19/56	0.6173 ± 0.2489 | NaN: 19/56	0.6333 ± 0.2299 | NaN: 20/56
Precision	0.6593 ± 0.2672 | NaN: 19/56	0.6796 ± 0.2492 | NaN: 20/56	0.6593 ± 0.2673 | NaN: 19/56	0.6588 ± 0.2671 | NaN: 19/56	0.6805 ± 0.2474 | NaN: 20/56
Recall	0.6652 ± 0.2165 | NaN: 21/56	0.6578 ± 0.2227 | NaN: 21/56	0.6652 ± 0.2165 | NaN: 21/56	0.6655 ± 0.2166 | NaN: 21/56	0.6606 ± 0.2177 | NaN: 21/56
Accuracy	0.9917 ± 0.0123 | NaN: 0/56	0.9917 ± 0.0123 | NaN: 0/56	0.9917 ± 0.0123 | NaN: 0/56	0.9917 ± 0.0123 | NaN: 0/56	0.9917 ± 0.0123 | NaN: 0/56
HD	101.7944 ± 84.8603 | NaN: 21/56	101.9575 ± 82.3769 | NaN: 21/56	101.7944 ± 84.8603 | NaN: 21/56	101.8023 ± 84.8868 | NaN: 21/56	102.7909 ± 84.4566 | NaN: 21/56
Count Error	4.3750 ± 5.5245 | NaN: 0/56	4.2321 ± 6.9872 | NaN: 0/56	4.3750 ± 5.5245 | NaN: 0/56	4.3214 ± 5.3556 | NaN: 0/56	3.9821 ± 5.8202 | NaN: 0/56
Steatosis Error	0.2222 ± 0.3260 | NaN: 0/56	0.2298 ± 0.3332 | NaN: 0/56	0.2222 ± 0.3261 | NaN: 0/56	0.2222 ± 0.3263 | NaN: 0/56	0.2269 ± 0.3325 | NaN: 0/56
	GT vs PRED				
empty-empty	19 (33.9%)	20 (35.7%)	19 (33.9%)	19 (33.9%)	20 (35.7%)
GT_empty-pred_pos	2 (3.6%)	1 (1.8%)	2 (3.6%)	2 (3.6%)	1 (1.8%)
GT_pos-pred_empty	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
GT_pos-pred_pos	35 (62.5%)	35 (62.5%)	35 (62.5%)	35 (62.5%)	35 (62.5%)
	MICRO - AVERAGED				
IoU	0.6735	0.6727	0.6735	0.6735	0.6731
Dice	0.8049	0.8043	0.8049	0.8049	0.8046
Precision	0.8297	0.8327	0.8296	0.8292	0.8321
Recall	0.7816	0.7778	0.7816	0.7820	0.7789
Accuracy	0.9917	0.9917	0.9917	0.9917	0.9917

Tab. 5 - FINAL RESULTS after the post-processing ablation study

6. Overall Segmentation Pipeline

The final pipeline (Fig. 10) is a streamlined three-step workflow. First, each H&E RGB image is stain-normalized using the Macenko method to reduce inter-slide color variability, without adding extra filtering or contrast manipulations. Second, the normalized image is passed to the selected CNN (SE-U-Net, Network 7A), which outputs a pixel-wise probability map for the target class. Finally, the probability map is binarized using the chosen threshold and refined with a single morphological opening step to suppress small isolated false positives and improve mask regularity.

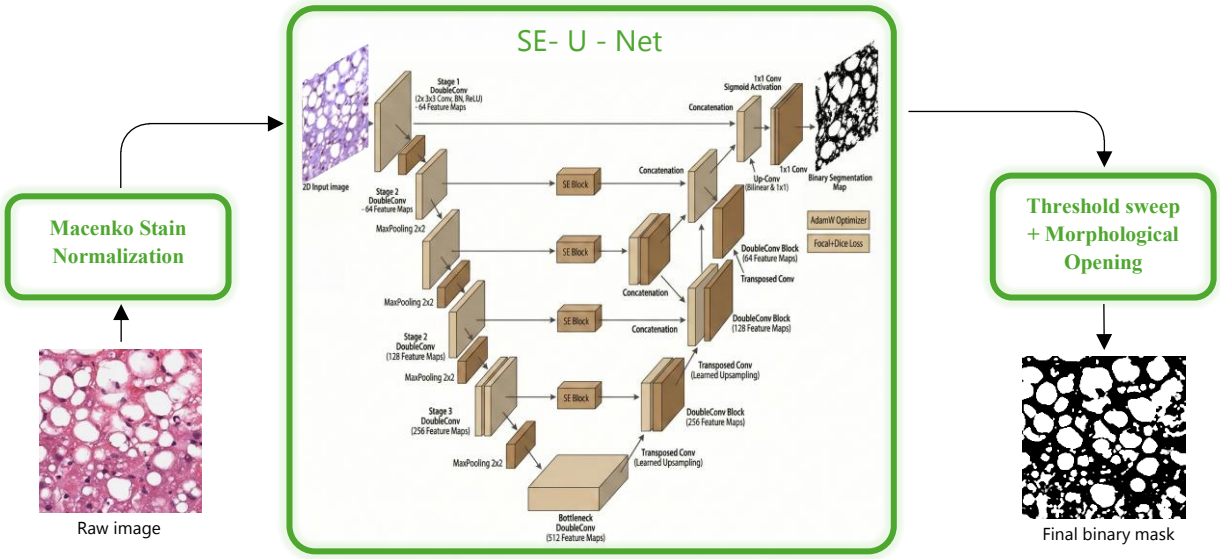


Fig. 10 - graphical representation of the final pipeline

7. Results and Discussion

The final pipeline was assessed on the Test Set and shows a solid and coherent behavior (Tab. 5), with stable overlap performance and a particularly contained error on global steatosis quantification. Importantly, the morphological opening step yields a small but consistent gain over the raw predictions, with the most visible impact at the instance level: by suppressing tiny, isolated detections, it reduces spurious components that would otherwise inflate the vacuole count. This is aligned with the intended role of opening—regularizing the mask without aggressively altering the main predicted regions.

From a real-world and clinical standpoint, aligning with fully automated quantitative assessment goals [3], two aspects deserve attention. First, the dataset contains a non-negligible fraction of images with no target structures, which makes some per-image overlap scores undefined and highlights a practical deployment issue: a clinically useful system should include an explicit “no-steatosis / no-vacuole” gate to avoid unnecessary false alarms and to make reporting robust across negative samples. Second, boundary quality and instance separation remain the main limitations, as vacuoles often appear clustered, partially fused, or weakly contrasted; under these conditions, connected component counting becomes sensitive to small artifacts and to under-segmentation.

Looking forward, the most meaningful developments would target robustness to staining and acquisition variability (e.g., multi-center generalization), and improved instance delineation—potentially through boundary-aware losses, instance-centric post-processing (e.g., watershed or marker-controlled separation), or complementary supervision focused on borders. These directions would be necessary to move from reliable global quantification toward more clinically actionable, per-vacuole measurements in routine workflows.

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